Total Pages: 02 Roll No……………………………..

Bangladesh University of Professionals (BUP)
Faculty of Business & Social Science (FBSS)
Department of Economics
76 LC 6th Semester Mid-Term Examination, October 2018

Course Code: ECON-3601 Course Title: Econometrics-II

Time: 3 Hours Full Marks: 100

Instructions:

a. Answer any five questions.
b. Each question carries equal Marks.
c. It is prohibited to use military abbreviation

---

1. a. Explain the following:
   i) Type-I & Type-II error; ii) Level of Significance iii) The p-value.

 b. Consider the following regression model and report the result of this analysis.

$$Y=24.45+0.50X$$
$$S_e= (6.41) \quad (0.03) \quad r^2 = 0.96$$
$$t = (3.81) \quad (14.26) \quad df=8$$
$$p = (0.003) \quad (0.0000003) \quad F_{1,8}=202.87$$

 c. State the concept histogram of residuals. 9+8+3=20

2. a. What is heteroscedasticity? Show that variance of the transformed disturbance term u* becomes homoscedastic and produce estimators that are BLUE.
   b. Distinguish between OLS and GLS. 10+10=20

3. a) Narrate the consequences of using OLS in the presence of heteroscedasticity.
   b) Proof that the presence of multicollinearity makes the estimators indeterminate.
   c) What is correlation matrix?- Explain. 8+8+4=20

4. a) Define auto correlation and classify it. Explain the motives for auto correlation.
   b) What do you know about The Breusch–Godfrey test-Explain. 10+10=20

5. a) Critically explain the Durbin-Watson d test for autocorrelation.
   b) What are the remedial measures for auto correlation? 16+4=20

6. a) What is specification errors? Classify it. What are the consequences of under fitting a model?
   b) Discuss the several steps of Ramsey's RESET test. 10+10=20

7. a) Mention the approaches using for developing a probability model for a binary response variable.
   b) Critically explain the Linear Probability Model.

8. a) What do you mean by dummy variables?-Illustrate.
   b) Use a hypothetical example and explain the cautions in the use of dummy variables.

4+16=20

9. Short notes (Any two) 5*2=10

a. ANOVA Table

b. Model selection criteria;

c. Logit model versus Probit model;

d. R^2 and the Adjusted R^2